



Large-Scale Time-Sensitive Networks
with
End-to-end Deterministic Service Requirements

Presentation at SARA, LAAS, Toulouse

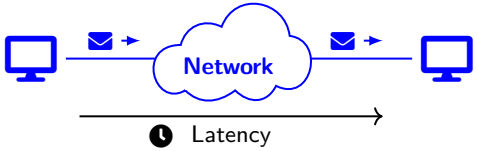
Ludovic Thomas

October 7th, 2026

In this Presentation:

1. Layer-2 Time-Sensitive Networks
2. DetNet: L3 Deterministic Networking
3. FRONTIER or The Need For Large-Scale Time-Sensitive Networks
4. Beyond DetNet: DNaaS
5. Conclusion

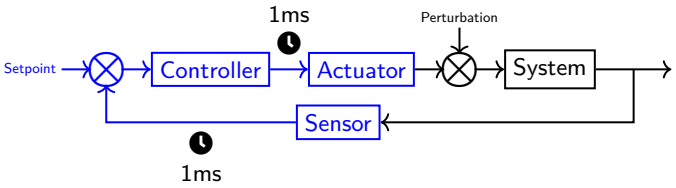


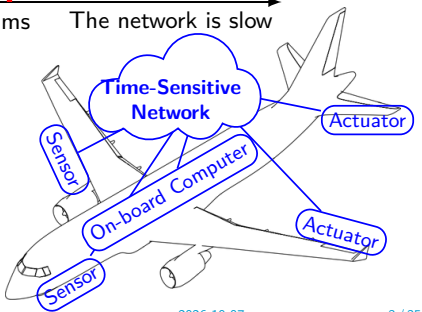








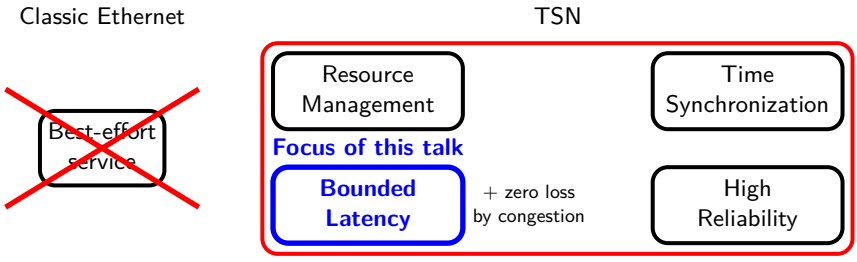




Cyber-physical Systems (CPS)

Time-Sensitive Networks

IEEE time-sensitive networking (TSN): the Ethernet for safety-critical applications (layer 2)

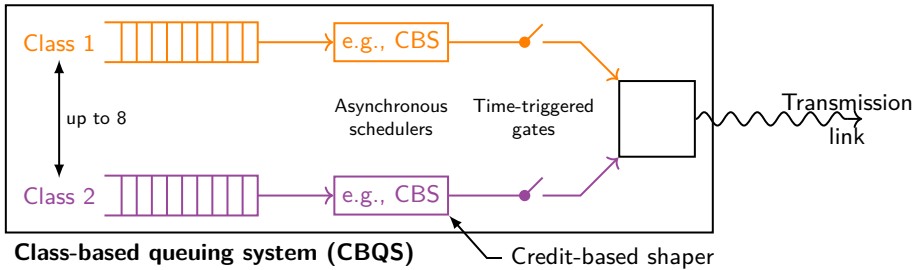


The four "key components" of TSN [**farkasTSNBasicConcepts2018a**]

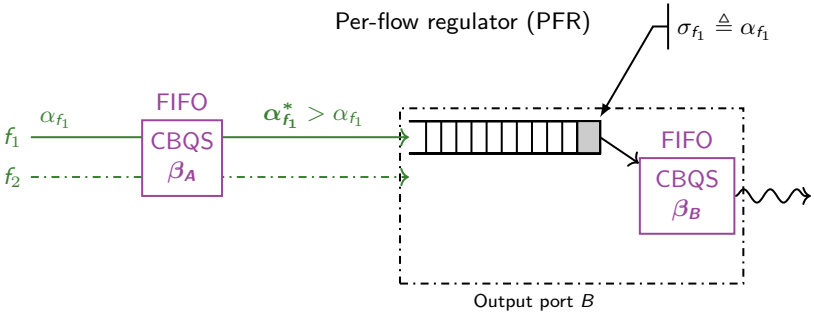
Similar ideas in **IETF deterministic networking (DetNet)** for IP and MPLS networks (layer 3) [RFC 8655]

– [Farkas 2018] János Farkas [Nov. 11, 2018]. "TSN Basic Concepts".
– [RFC 8655] Norman Finn, Pascal Thubert, Balazs Varga, and János Farkas [Oct. 2019]. *Deterministic Networking Architecture*. Request for Comments RFC 8655. Internet Engineering Task Force. DOI: 10.17487/RFC8655

Class-Based Queuing Mechanisms improve the Classic Ethernet Output Port

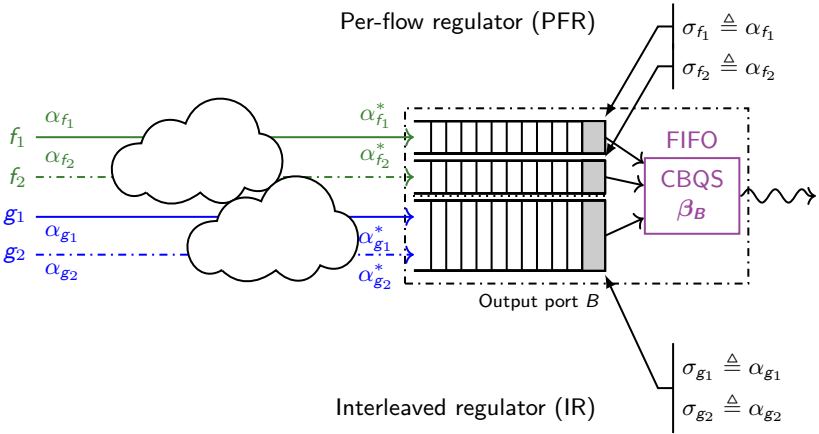


Within One Class, **Traffic Regulators** Allow for a Per-Flow Control



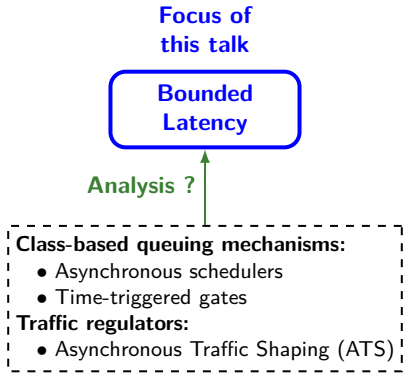
CBQS: Class-based queuing system (FIFO for the class aggregate)

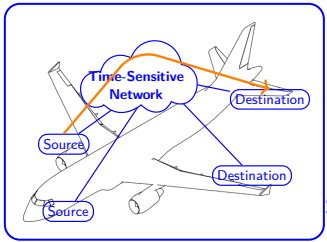
Within One Class, **Traffic Regulators** Allow for a Per-Flow Control



TSN *Asynchronous Traffic Shaping* (ATS)

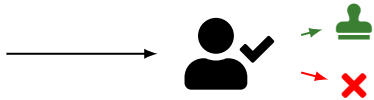
CBQS: Class-based queuing system (FIFO for the class aggregate)


Focus of
this talk
Bounded
Latency
Analysis ?
Class-based queuing mechanisms:
• Asynchronous schedulers
• Time-triggered gates
Traffic regulators:
• Asynchronous Traffic Shaping (ATS)



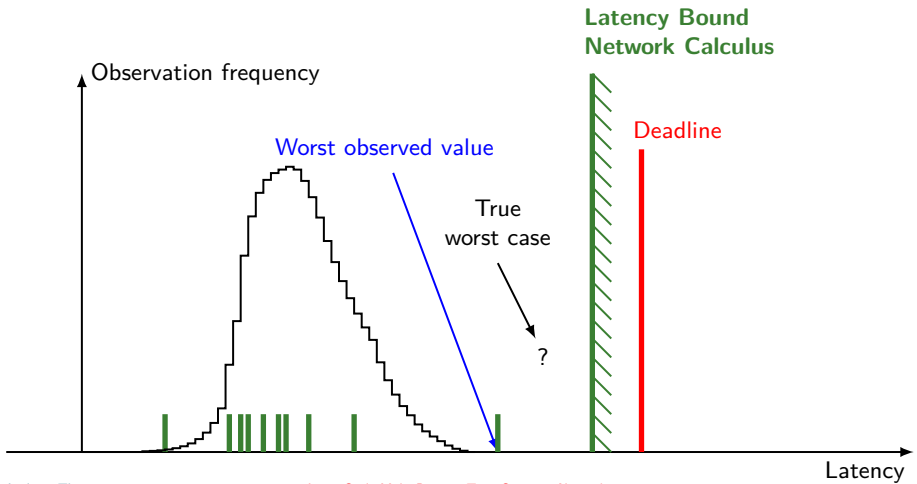


System owner
(ex: Airbus)



Certification authority
(ex: European Aviation Safety Agency)




Latency Bound
Network Calculus
Observation frequency
Deadline
Worst observed value
True
worst case
?
Latency

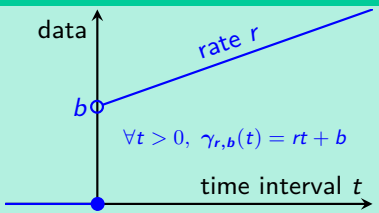
Network Calculus Provides a **Model-Based** Analysis and **Verification**

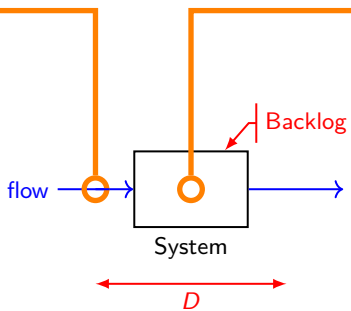
Two examples of models:

Arrival Curve $\alpha(t)$

upper-bounds the **maximum amount of traffic** of the flow over any interval

Leaky-Bucket $\gamma_{r,b}$

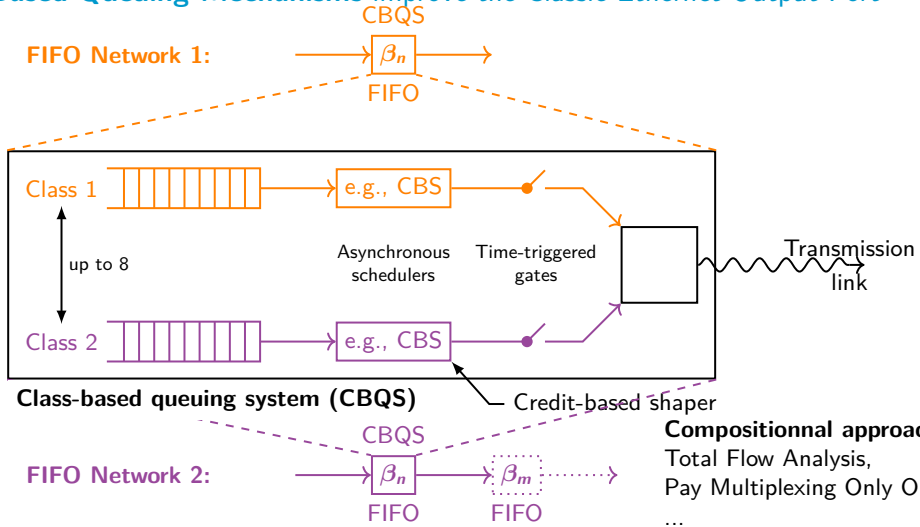




Service Curve $\beta(t)$

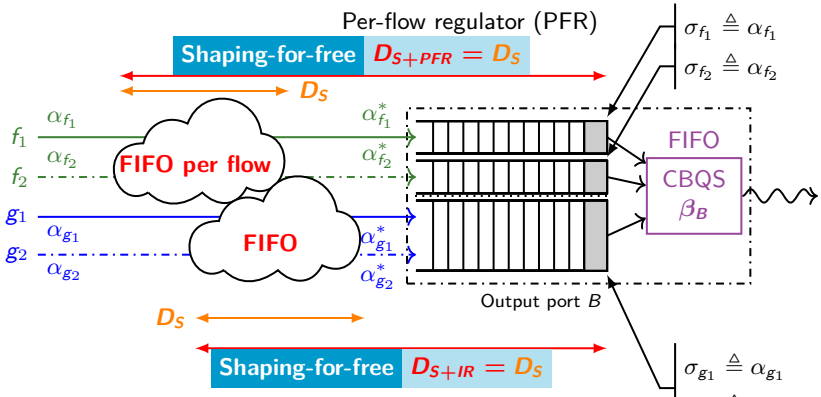
captures the **minimum service** guaranteed to the flow by the system over any interval

Class-Based Queuing Mechanisms improve the Classic Ethernet Output Port



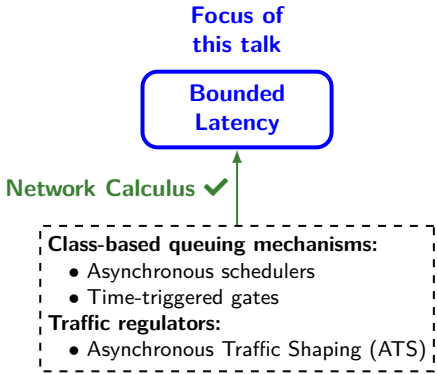
Compositionnal approaches:
Total Flow Analysis,
Pay Multiplexing Only Once,
...

Within One Class, **Traffic Regulators** Allow for a Per-Flow Control



TSN *Asynchronous Traffic Shaping* (ATS)

CBQS: Class-based queuing system (FIFO for the class aggregate)


Focus of this talk
Bounded Latency
Network Calculus ✔
Class-based queuing mechanisms:
• Asynchronous schedulers
• Time-triggered gates
Traffic regulators:
• Asynchronous Traffic Shaping (ATS)

Conclusion: **Status of L2 Time-Sensitive Networks**

- The A380, the A400M, and the A350 fly with AFDX networks, certified with Network Calculus.
- IEEE TSN is being deployed in Industry Automation, In-Vehicle Networks, etc.
- Network Calculus is versatile enough to also analyze other L2 time-sensitive networks: 5G, etc

Remaining Issues

- IEEE TSN is complex, industrials need easy-to-use configuration tools.
- Some TSN mechanisms need to be amended to fix issues identified by academia.
- Most industrials only need a fraction of IEEE TSN's capabilities.
- Need for interactions between technologies (TSN - 5G).

Need a national experimental platform to test configurations & new or modified mechanisms ?

Join SCLICES-FR's *thematic group for expressing needs in time-sensitive networks* (GTEB-temps réel).

1 Layer-2 Time-Sensitive Networks

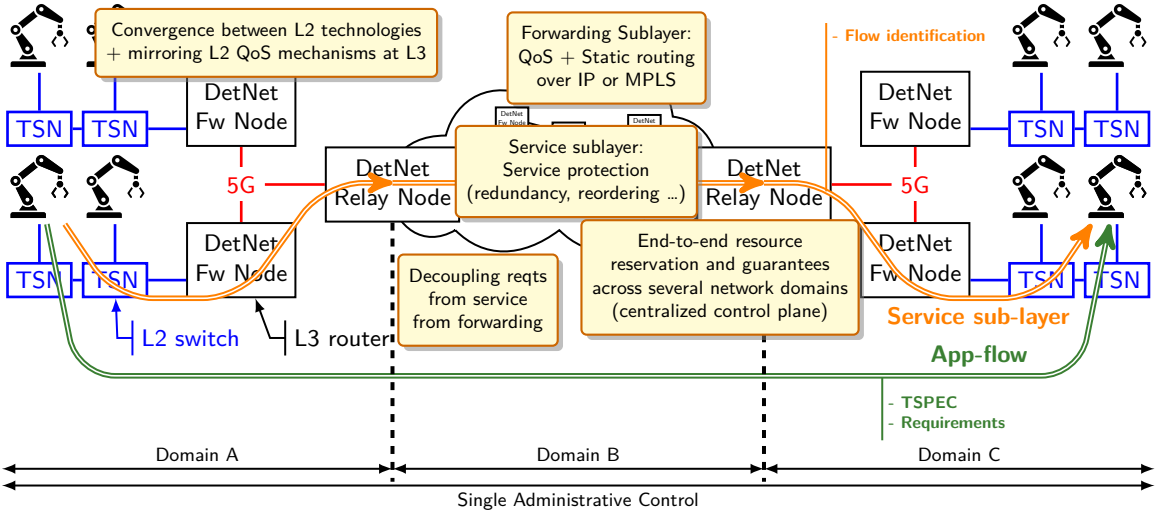
2 DetNet: L3 Deterministic Networking

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4 Beyond DetNet: DNaaS

5 Conclusion

DetNet: L3 Deterministic Networking



Icons from flaticon

DetNet: L3 Deterministic Networking

Use-cases are available in [RFC 8578].

Excerpt from DetNet's charter

The Working Group will initially focus on solutions for networks that are under a single administrative control or within a closed group of administrative control; these include not only campus-wide networks but also private WANs.

Campus-wide networks are not very "large", how about nationwide or continental networks?

– [RFC 8578] Ethan Grossman [May 2019]. *Deterministic Networking Use Cases*. Request for Comments RFC 8578. Internet Engineering Task Force. DOI: 10.17487/RFC8578

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The Needs

Why do we not need large-scale time sensitive networks?

Because physical phenomena have a maximum propagation speed (a plane has a maximum speed, we don't need to connect its take-off and landing airports with time-sensitive networks).

But there is at least one CPS where physical phenomena propagate at the speed of light: the **Electrical Grid**.

The Continental Europe Synchronous Area (CESA)



Figure: Continental Europe Synchronous Area. From ENTSOE.

The CESA is an AC grid with nominal frequency of **50 Hz**. The grid is stable (remains ∼50Hz) if at any point in time, the **power generated equals the power consumed**.

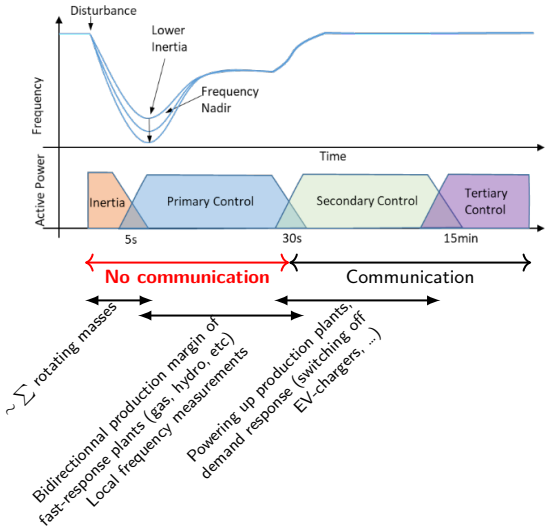
Overproduction → frequency increases.
Underproduction → frequency decreases.

Frequency must remain within ±0.2 Hz of the nominal frequency, and electrical equipments start to disconnect if the frequency goes beyond ±1 Hz.
See `https://mainsfrequency.com/`.

NB: Since March 16th, 2022, Ukraine is also connected to CESA.

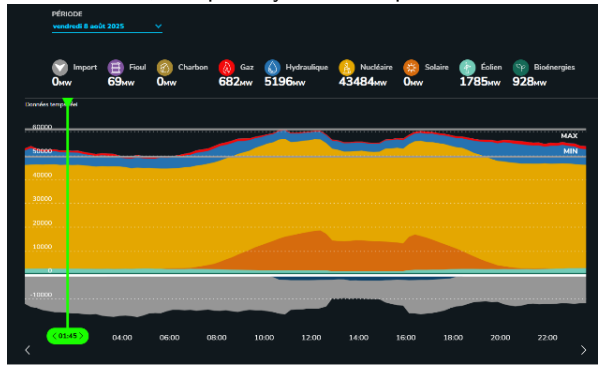
Grid Stability 101

Today's Stability Principles:



Two Upcoming Issues:

⇒ Higher share of intermittent renewable energy sources
→ less inertia, less primary-control capabilities.



Curtailment of renewable energy sources in summer.

⇒ Primary control does not consider powerline limits.

Grid Stability in the Future

Futurs Énergétiques 2050 - Principaux Résultats. [RTE 2022]

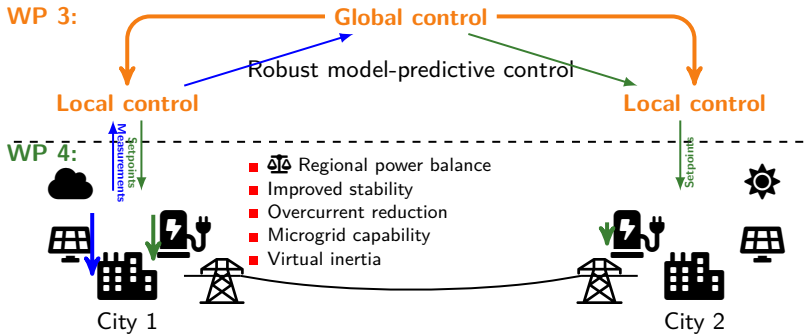
- "*All scenarios towards carbon neutrality in 2050 require the deployment of demand-response flexibilities*" (e.g., EV chargers)
- Need for services of fast frequency response (e.g., based on demand response)
- "*The role of digital technologies will be increasingly important in the real-time management of a complex electrical system composed of more dispersed and weather-dependent generation resources*"

– [RTE 2022] RTE [Feb. 2022]. *Futurs Énergétiques 2050 - Rapport Complet*.

FRONTIER (PEPR Future Networks)

Future real-time networks towards the balance of large-scale cyber-physical systems and their optimization.

LORIA
ISAE-SUPAERO
LIP6
IRISA
G2ELAB



FRONTIER (PEPR Future Networks)

Recall from DetNet's Charter

Excerpt from DetNet's charter

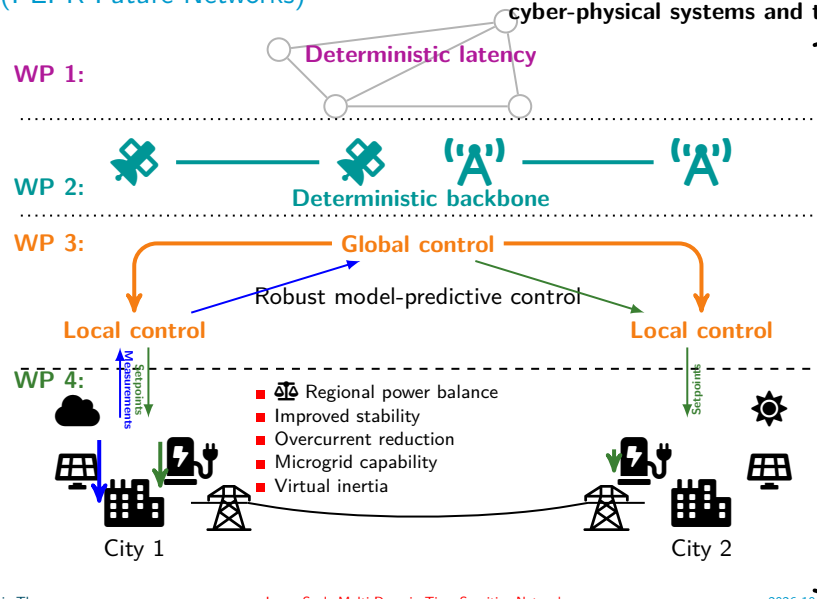
*The Working Group will initially focus on solutions for networks that are under a **single administrative control or within a closed group of administrative control**; these include not only campus-wide networks but also private WANs.*

How do we reach a continental scale with remotely-accessible devices (offshore windturbines, etc) within a single administrative control ?

FRONTIER (PEPR Future Networks)

Future real-time networks towards the balance of large-scale cyber-physical systems and their optimization.

LORIA
ISAE-SUPAERO
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IRISA
G2ELAB



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DetNet's Strongest Limitation

Excerpt from DetNet's charter

The Working Group will initially focus on solutions for networks that are under a ***single administrative control or within a closed group of administrative control****; these include not only campus-wide networks but also private WANs.* ***The DetNet WG will not spend energy on solutions for large groups of domains such as the Internet.***

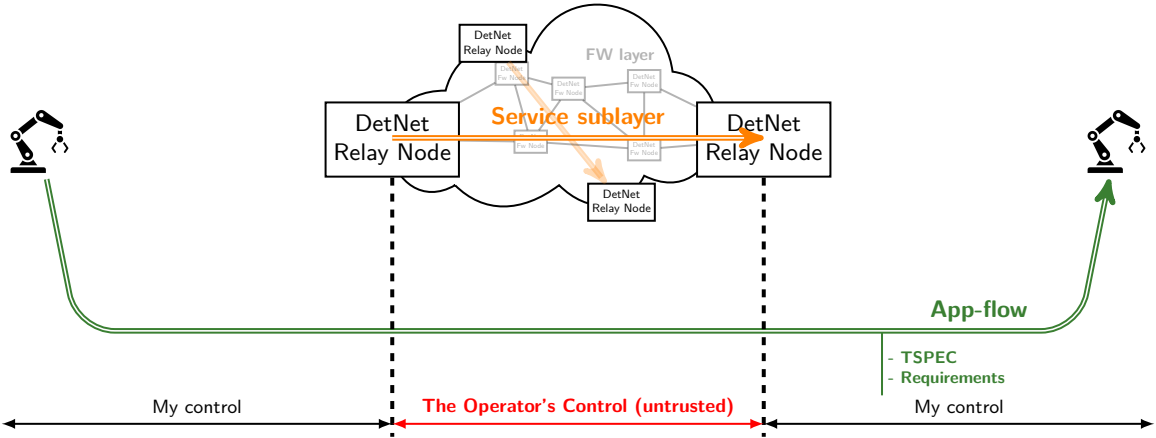
That's too bad....

But we shall not forget why ATM, IntServ, etc. failed at being deployed at large scale

- Multiple administrative entities **do not trust each other** and **do not want to share their network's configuration**.
- The best-effort model of the Internet allows for interconnecting trustless networks.
- On the Internet, contractual agreements (SLAs) can be verified through monitoring stochastic quantities (means, 95% values etc.) $\Rightarrow$ this is not sufficient for deterministic service agreements (DSA).

Can we find a solution for deterministic service agreements (DSA)?

Trustless Deterministic Networking with **Customer Privacy** and **Confidential Configuration**



Network calculus can check the deterministic requirements but it requires the complete knowledge of the traffic and the network configuration. Only the operator has this knowledge. I (as a client) don't trust the operator.

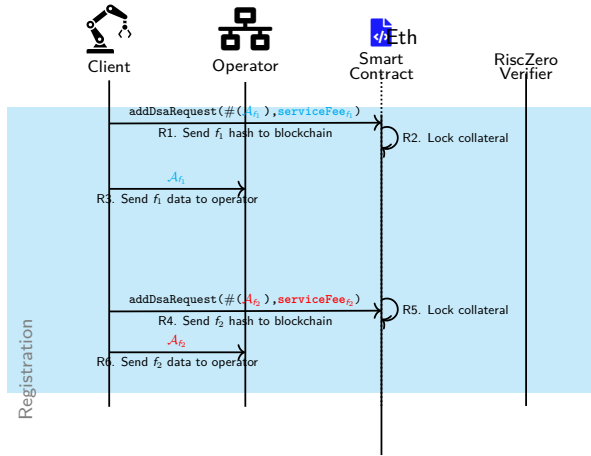
Pieces of solution: Zero-knowledge network-calculus proofs and smart contracts.

Icons from flaticon

DNaaS: Deterministic Network as a Service [Amet, Thomas, Song 2026]



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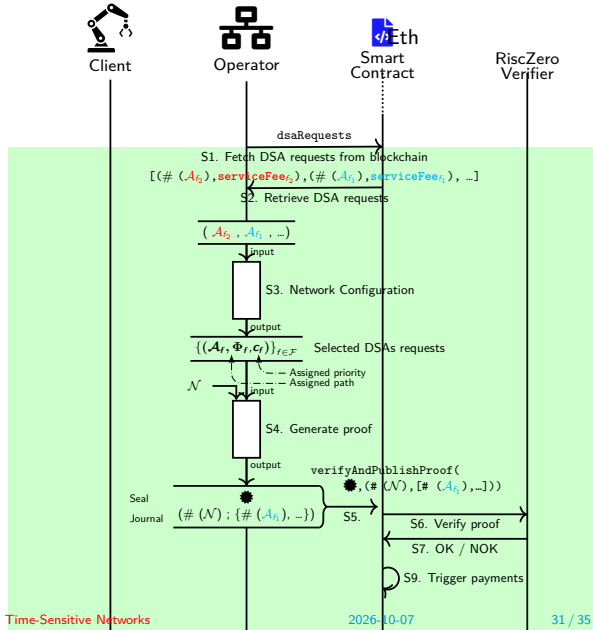


– [Amet, Thomas, Song 2026] Matthieu Amet, Ludovic Thomas, and Ye-Qiong Song [May 2026]. "DNaaS: Towards Scalable and Trusted Time-Sensitive Networks". In: *IEEE INFOCOM 2026 - IEEE Conference on Computer Communications*. DOI: 10.1109/INFOCOM59046.2026.11571313

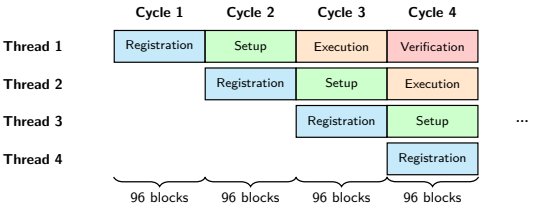
DNaaS: Deterministic Network as a Service



...



DNaaS: Deterministic Network as a Service



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Conclusion

- There is a need for large-scale time-sensitive networks
- DetNet partially addresses this need
- DNaaS aims at going beyond the current charter of DetNet

Bibliography I

[Amet, Thomas, Song 2026] Amet, Matthieu, Ludovic Thomas, and Ye-Qiong Song (May 2026). "DNaaS: Towards Scalable and Trusted Time-Sensitive Networks". In: *IEEE INFOCOM 2026 - IEEE Conference on Computer Communications*. IEEE INFOCOM 2026 - IEEE Conference on Computer Communications, pp. 1–10. DOI: 10.1109/INFOCOM59046.2026.11571313. URL: https://ieeexplore.ieee.org/document/11571313 (visited on 07/09/2026).

[Farkas 2018] Farkas, János (Nov. 11, 2018). "TSN Basic Concepts". URL: https://www.ieee802.org/1/files/public/docs2018/detnet-tsn-farkas-tsn-basic-concepts-1118-v01.pdf.

[Finn, *et al.* 2019] Finn, Norman et al. (Oct. 2019). *Deterministic Networking Architecture*. Request for Comments RFC 8655. Internet Engineering Task Force. 38 pp. DOI: 10.17487/RFC8655. URL: https://datatracker.ietf.org/doc/rfc8655 (visited on 11/09/2022).

[RTE 2022] RTE (Feb. 2022). *Futurs Énergétiques 2050 - Rapport Complet*. URL: https://assets.rte-france.com/prod/2022-06/Futurs%20%C3%A9nerg%C3%A9tiques%202050%20_%20rapport%20complet.zip (visited on 12/10/2022).

[RFC 8655] Finn, Norman et al. (Oct. 2019). *Deterministic Networking Architecture*. Request for Comments RFC 8655. Internet Engineering Task Force. 38 pp. DOI: 10.17487/RFC8655. URL: https://datatracker.ietf.org/doc/rfc8655 (visited on 11/09/2022).

Bibliography II

[RFC 8578] Grossman, Ethan (May 2019). *Deterministic Networking Use Cases*. Request for Comments RFC 8578. Internet Engineering Task Force. 97 pp. DOI: 10.17487/RFC8578. URL: https://datatracker.ietf.org/doc/rfc8578 (visited on 11/06/2022).